Comparison of Halo Concentration Predictions between Planck and Other Cosmologies

Description

In this supplementary material, we present a systematic comparison of halo concentrations predicted by our universal concentration model (from $\Delta_{\rm h} = \Delta_{\rm vir}$) for a broad range of cosmological models over the redshift interval $z = 0$–2. Throughout this analysis, Planck18 is adopted as the reference cosmology, and the predicted concentrations in other cosmologies are compared against it.

The cosmological models considered here are intended to cover as broad a range as possible, including several commonly used ΛCDM parameter sets, as well as cosmologies with different values of σ_8, $\Omega_{\rm m}$, w, $m_{\rm WDM}$, and $n_{\rm s}$. The cosmological parameters are listed in Table 1. The corresponding $c_{\rm vir}$–$M_{\rm vir}$ relations are shown in Fig. 1, while Table 2 summarizes the relative concentration differences, $\Delta c/c$, at several representative halo masses.

Results

The main results can be summarized as follows:

1. Among the standard cosmologies considered here, Planck18 generally predicts higher concentrations than the WMAP cosmologies, except for WMAP1. The overall ordering of concentration amplitude is

   $$\text{WMAP3} < \text{WMAP5} < \text{WMAP7} < \text{WMAP9} < \text{Planck18} < \text{WMAP1}.$$

   We have also tested several Planck cosmologies and find that their predicted concentrations are nearly identical over the mass and redshift ranges considered here. Therefore, only Planck18 is shown explicitly.

2. The concentration predictions for Planck18 are very close to those for MultiDark-Planck, and both are systematically higher than those for Illustris and Bolshoi. The Millennium cosmology yields the highest concentrations among the examples shown. Overall, the concentration level depends primarily on the cosmological parameters σ_8 and $\Omega_{\rm m}$ (see Table 1 and Fig. 1). A more detailed discussion of the effects of varying σ_8, $\Omega_{\rm m}$, w, $m_{\rm WDM}$, and $n_{\rm s}$ is provided in the main text.

3. For most cosmological variations, the relative concentration differences become smaller toward higher halo masses and also tend to decrease at higher redshift.

Remarks on the ΛWDM Comparison

The comparison between ΛWDM cosmologies and Planck18 deserves additional clarification. At low halo masses, ΛWDM predicts significantly lower concentrations than Planck18. This behavior is physically expected, because the free-streaming of warm dark matter suppresses the formation of small-scale density perturbations, delays halo formation, and thus lowers the concentrations of low-mass halos.

At high halo masses, the physical effect of warm-dark-matter free-streaming should become negligible, and the ΛWDM concentration should approach that of the corresponding CDM cosmology. Therefore, in principle, the difference between ΛWDM and Planck18 is expected to be very small in the massive-halo regime.

In our comparison, however, a residual difference remains even at high masses. This offset arises mainly from the different window functions adopted in the halo mass variance calculation: for ΛWDM cosmologies, we use a sharp-k filter, while for Planck18 we use a real-space top-hat filter. Since these two filters do not lead to exactly the same variance–mass correspondence, they introduce a systematic difference in the predicted concentrations. Accordingly, the large discrepancy at low masses should be interpreted as the genuine physical suppression caused by warm-dark-matter free-streaming, whereas the residual difference at high masses is primarily a methodological effect associated with the different window-function choices.

Table 1: Pre-set ΛCDM cosmologies.

ID	H_0	$\Omega_{\rm m}$	$\Omega_{\rm b}$	$n_{\rm s}$	σ_8	Reference
Planck18	67.66	0.3111	0.0490	0.9665	0.8102	Planck Collaboration et al. [2020]
Planck15	67.74	0.3089	0.0486	0.9667	0.8159	Planck Collaboration et al. [2016]
Planck13	67.77	0.3071	0.0483	0.9611	0.8288	Planck Collaboration et al. [2014]
WMAP9	69.32	0.2865	0.0463	0.9608	0.8200	Hinshaw et al. [2013]
WMAP7	70.20	0.2743	0.0458	0.9680	0.8160	Komatsu et al. [2011]
WMAP5	70.50	0.2732	0.0456	0.9600	0.8120	Komatsu et al. [2009]
WMAP3	73.50	0.2342	0.0413	0.9510	0.7420	Spergel et al. [2007]
WMAP1	72.00	0.2700	0.0463	0.9900	0.9000	Spergel et al. [2003]
Illustris	70.40	0.2726	0.0456	0.9630	0.8090	Vogelsberger et al. [2014]
Bolshoi	70.00	0.2700	0.0469	0.9500	0.8200	Klypin et al. [2011]
Multidark-Planck	67.80	0.3070	0.0480	0.9600	0.8290	Klypin et al. [2016]
Millennium	73.00	0.2500	0.0450	1.0000	0.9000	Springel et al. [2005]

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Table 2: The $\Delta c/c$ for the Planck18 and other cosmologies at $z = 0.0 \sim 2.0$.

Cosmology	Halo Mass ($h^{-1}M_\odot$)																							
	1×10^8			1×10^9			1×10^{10}			1×10^{11}			1×10^{12}			1×10^{13}			1×10^{14}			1×10^{15}		
	$z=0$	$z=1$	$z=2$	$z=0$	$z=1$	$z=2$	$z=0$	$z=1$	$z=2$	$z=0$	$z=1$	$z=2$	$z=0$	$z=1$	$z=2$	$z=0$	$z=1$	$z=2$	$z=0$	$z=1$	$z=2$	$z=0$	$z=1$	$z=2$
WMAP9	−0.03	−0.01	−0.00	−0.03	−0.01	−0.00	−0.03	−0.00	0.00	−0.02	−0.00	0.00	−0.02	0.00	0.00	−0.01	0.00	0.00	−0.01	0.01	0.01	−0.00	0.01	0.00
WMAP7	−0.05	−0.01	−0.00	−0.04	−0.01	−0.00	−0.04	−0.01	0.00	−0.04	−0.00	0.00	−0.03	0.00	0.00	−0.02	0.00	0.00	−0.02	0.01	0.01	−0.01	0.01	0.01
WMAP5	−0.06	−0.02	−0.01	−0.06	−0.02	−0.01	−0.05	−0.02	−0.01	−0.05	−0.01	−0.00	−0.04	−0.01	−0.00	−0.03	−0.00	0.00	−0.02	0.00	0.00	−0.01	0.01	0.00
WMAP3	−0.21	−0.12	−0.09	−0.20	−0.11	−0.07	−0.19	−0.10	−0.06	−0.17	−0.08	−0.05	−0.15	−0.06	−0.03	−0.13	−0.05	−0.02	−0.10	−0.03	−0.01	−0.07	−0.01	−0.00
WMAP1	0.10	0.12	0.10	0.09	0.11	0.09	0.08	0.10	0.08	0.08	0.09	0.07	0.07	0.08	0.05	0.06	0.06	0.04	0.05	0.05	0.03	0.04	0.04	0.02
Illustris	−0.06	−0.02	−0.01	−0.06	−0.02	−0.01	−0.05	−0.02	−0.01	−0.05	−0.01	−0.00	−0.04	−0.01	−0.00	−0.03	−0.00	0.00	−0.02	0.00	0.00	−0.01	0.01	0.00
Bolshoi	−0.08	−0.04	−0.03	−0.08	−0.03	−0.02	−0.07	−0.02	−0.01	−0.06	−0.02	−0.01	−0.04	−0.01	−0.00	−0.03	−0.00	0.00	−0.02	0.01	0.01	−0.00	0.01	0.01
MultiDark-Planck	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.00
Millennium	0.08	0.12	0.11	0.07	0.11	0.09	0.06	0.10	0.08	0.05	0.09	0.07	0.05	0.07	0.06	0.04	0.06	0.05	0.04	0.05	0.03	0.04	0.04	0.02
$\sigma_8 = 0.7$	−0.14	−0.11	−0.09	−0.13	−0.11	−0.08	−0.13	−0.10	−0.07	−0.12	−0.09	−0.06	−0.12	−0.08	−0.05	−0.11	−0.07	−0.04	−0.09	−0.05	−0.03	−0.08	−0.04	−0.02
$\sigma_8 = 0.8$	−0.01	−0.01	−0.01	−0.01	−0.01	−0.01	−0.01	−0.01	−0.01	−0.01	−0.01	−0.01	−0.01	−0.01	−0.01	−0.01	−0.01	−0.00	−0.01	−0.01	−0.00	−0.01	−0.00	−0.00
$\sigma_8 = 0.9$	0.11	0.09	0.07	0.11	0.09	0.07	0.11	0.08	0.06	0.10	0.08	0.05	0.10	0.07	0.04	0.09	0.06	0.04	0.08	0.05	0.03	0.06	0.03	0.02
$\sigma_8 = 1.0$	0.24	0.20	0.16	0.24	0.19	0.15	0.23	0.18	0.13	0.22	0.16	0.11	0.21	0.15	0.10	0.19	0.12	0.08	0.17	0.10	0.06	0.13	0.07	0.04
$\omega_m = 0.2$	−0.23	−0.12	−0.07	−0.22	−0.10	−0.06	−0.20	−0.08	−0.04	−0.18	−0.06	−0.03	−0.15	−0.04	−0.01	−0.11	−0.01	0.00	−0.07	0.01	0.01	−0.02	0.03	0.02
$\omega_m = 0.3$	−0.02	−0.01	−0.01	−0.02	−0.01	−0.01	−0.02	−0.01	−0.00	−0.02	−0.01	−0.00	−0.01	−0.00	−0.00	−0.01	−0.00	−0.00	−0.01	0.00	0.00	−0.00	0.00	0.00
$\omega_m = 0.4$	0.19	0.09	0.06	0.17	0.07	0.05	0.16	0.06	0.04	0.14	0.05	0.02	0.11	0.03	0.01	0.08	0.01	0.00	0.05	−0.00	−0.00	0.02	−0.01	−0.01
$\omega_m = 0.5$	0.40	0.18	0.12	0.37	0.16	0.10	0.33	0.13	0.08	0.29	0.10	0.05	0.24	0.06	0.03	0.18	0.03	0.01	0.11	0.00	−0.01	0.04	−0.02	−0.01
$w = -0.7$	0.00	0.06	0.08	0.00	0.05	0.07	0.00	0.05	0.07	0.00	0.05	0.06	0.00	0.04	0.05	−0.00	0.04	0.04	0.00	0.03	0.03	−0.00	0.02	0.02
$w = -0.9$	0.00	0.02	0.02	0.00	0.02	0.02	0.00	0.02	0.02	0.00	0.01	0.02	0.00	0.01	0.01	0.00	0.01	0.01	0.00	0.01	0.01	−0.00	0.01	0.00
$w = -1.1$	−0.00	−0.02	−0.02	−0.00	−0.01	−0.02	−0.00	−0.01	−0.02	0.00	−0.01	−0.01	0.00	−0.01	−0.01	0.00	−0.01	−0.01	−0.00	−0.01	−0.01	0.00	−0.01	−0.00
$w = -1.3$	−0.00	−0.04	−0.05	−0.00	−0.04	−0.04	−0.00	−0.04	−0.04	−0.00	−0.04	−0.04	−0.00	−0.03	−0.03	−0.00	−0.03	−0.02	−0.00	−0.02	−0.02	0.00	−0.02	−0.01
$m_{\rm WDM} = 0.5$ keV	−0.84	−0.70	−0.55	−0.82	−0.66	−0.50	−0.78	−0.61	−0.44	−0.60	−0.44	−0.31	0.13	0.09	0.06	0.57	0.37	0.23	0.54	0.32	0.18	0.42	0.22	0.11
$m_{\rm WDM} = 1.0$ keV	−0.84	−0.70	−0.55	−0.80	−0.65	−0.49	−0.62	−0.48	−0.35	0.15	0.11	0.08	0.63	0.44	0.29	0.64	0.42	0.26	0.56	0.32	0.19	0.42	0.22	0.11
$m_{\rm WDM} = 1.5$ keV	−0.83	−0.69	−0.54	−0.75	−0.61	−0.46	−0.23	−0.18	−0.13	0.54	0.40	0.28	0.69	0.48	0.32	0.65	0.42	0.26	0.56	0.32	0.19	0.42	0.22	0.11
$m_{\rm WDM} = 2.0$ keV	−0.82	−0.68	−0.53	−0.63	−0.51	−0.39	0.15	0.12	0.09	0.67	0.49	0.34	0.71	0.49	0.33	0.65	0.42	0.26	0.56	0.32	0.19	0.42	0.22	0.11
$n_s = 1.0$	21816.56	12571.69	9278.96	4449.48	2480.78	1772.10	931.46	498.24	342.30	200.88	101.86	66.78	44.58	21.02	13.02	9.96	4.20	2.43	2.02	0.67	0.35	0.19	−0.02	−0.02
$n_s = 0.0$	1040.86	599.58	442.51	332.07	184.94	132.08	108.44	57.83	39.71	36.24	18.22	11.93	12.29	5.68	3.50	4.11	1.65	0.94	1.25	0.37	0.19	0.26	0.01	−0.00
$n_s = -1.0$	48.87	27.94	20.59	24.06	13.20	9.40	11.98	6.21	4.24	6.01	2.87	1.86	3.01	1.26	0.76	1.48	0.50	0.28	0.71	0.16	0.08	0.34	0.03	0.01
$n_s = -1.5$	10.00	5.54	4.06	5.96	3.11	2.19	3.56	1.70	1.14	2.12	0.89	0.56	1.27	0.43	0.25	0.77	0.19	0.10	0.50	0.08	0.03	0.38	0.05	0.02
$n_s = -2.0$	1.50	0.64	0.44	1.01	0.35	0.22	0.68	0.16	0.08	0.46	0.05	0.01	0.33	−0.01	−0.02	0.29	−0.02	−0.03	0.32	0.01	−0.00	0.43	0.06	0.02

Note. Negative (positive) values indicate that the predicted concentration for a given cosmology is lower (higher) than that of Planck18 by $\Delta c/c$.

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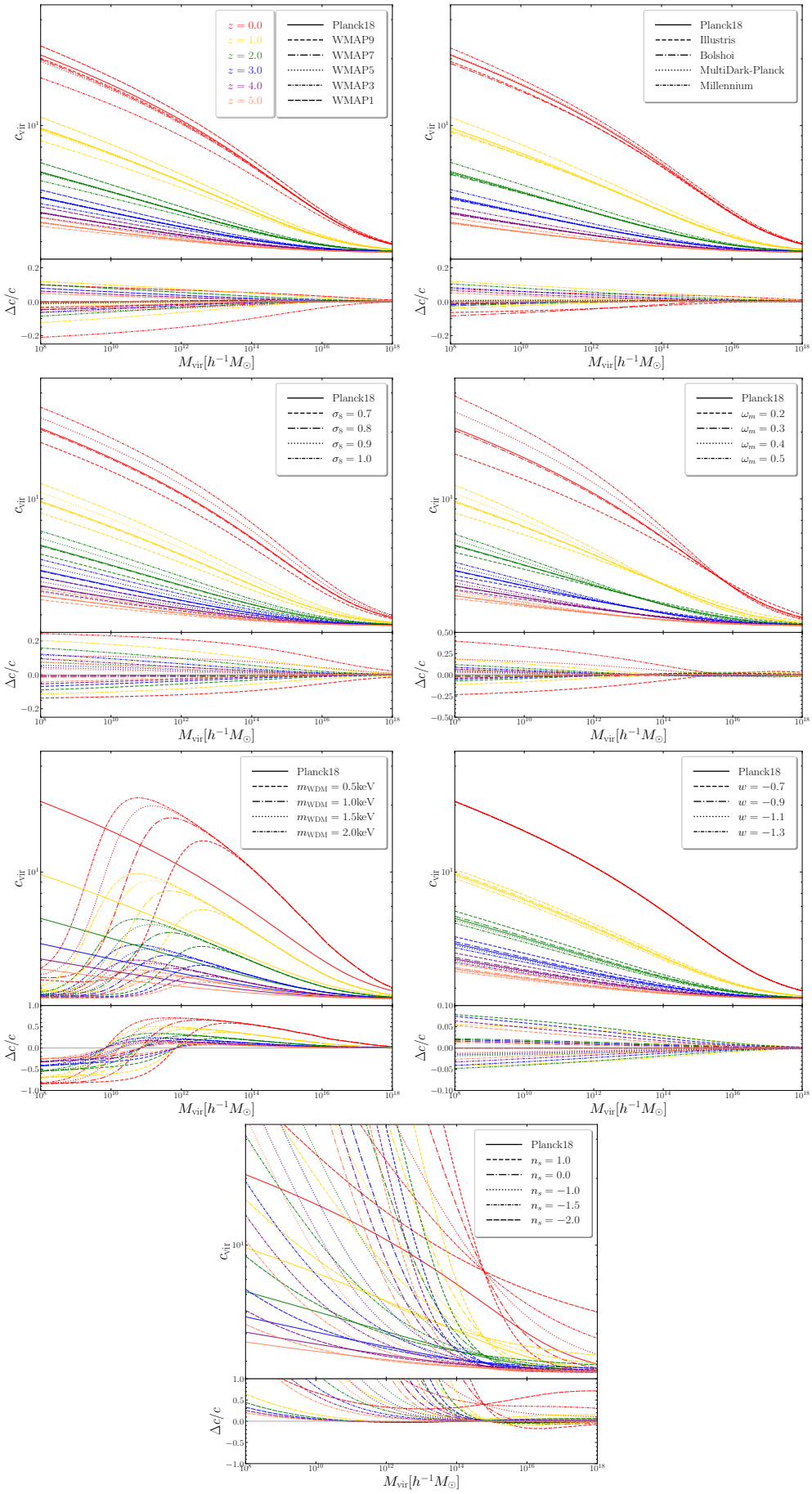

Figure 1: Predicted $c_{\rm vir}$–$M_{\rm vir}$ relations for different cosmological models.